Nurun Nahar Fiha

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Education

Jahangirnagar University **Savar, Bangladesh**
M.Sc. in Computer Science and Engineering *Jun 2024 – Sep 2025*

- **CGPA: 3.83/4.00 (Rank: 7/44)**
- **Notable Courses:** Natural Language Processing, Machine Intelligence & Expert Systems, Network Security, Distributed Database, Pattern Recognition

B.Sc. in Computer Science and Engineering *Feb 2019 – May 2024*

- **CGPA: 3.69/4.00 (Rank: 9/59)**
- **Notable Courses:** Machine Learning, AI, Data Structures & Algorithms, Database Systems, Operating Systems, HCI

Research Experience

Multilingual Sentiment Analysis of Social Media Reactions to OpenAI's Ghibli-Style Image Generation Feature

[Manuscript in preparation] — M.Sc. thesis; supervised by Dr. Israt Jahan

- Built a 36.7K-record multilingual corpus of Twitter and YouTube reactions to a viral generative-AI feature to study public sentiment toward AI-generated art, using machine translation of non-English posts, emoji-aware preprocessing, and VADER-based weak supervision.
- Benchmarked 21 transformer and hybrid Transformer-RNN configurations on accuracy and calibration (ECE). BERT-base performed best (91.27% accuracy on VADER-derived labels), while a lighter 1-layer DistilBERT-BiGRU hybrid nearly matched it (91.12%) with the lowest calibration error of all models.
- Found that accuracy varied by under 3 percentage points across models while calibration error varied more than fourfold, and that 16 of 18 hybrids were worse calibrated than their base transformer.

Deep Learning Based Bangladeshi Currency Coin Recognition

[Published at ICREST 2025, IEEE Xplore] — B.Sc. thesis; supervised by Md. Golam Moazzam

- Addressed coin identification for visually impaired users, as Bangladeshi coins are nearly identical in size and texture, by building a 2,800-image, 14-class dataset captured under real-world lighting and rotation conditions.
- Trained a Faster R-CNN (ResNet-50 FPN) against a CNN baseline in PyTorch, reaching 91.18% test accuracy (vs. 84.04%).

A Comparative Analysis of Low-Resource Techniques for Named Entity Recognition in Bengali

M.Sc. course research project (NLP)

- Investigated the effectiveness of modeling approaches for Bengali, a widely spoken but low-resource language, by benchmarking CRF, BiLSTM-CRF, mBERT, and XLM-RoBERTa on a custom 25K-sentence dataset merged from three public corpora.
- Enhanced models with FastText embeddings, character-level CNNs, and linguistic features; XLM-RoBERTa achieved the best F1 (0.758).

Cyberbullying Detection on Social Media Platforms Using Machine Learning

B.Sc. course research project (ML)

- Relabeled a 16K-comment Bengali toxic-comment dataset into 3 classes (Non-Bullying, Bullying, Crime) and compared four classifiers (LR, DT, RF, XGBoost). XGBoost with class-imbalance weighting performed best, with 75.4% test accuracy and a 0.761 10-fold CV score.

Awards & Recognition

National Science and Technology (NST) Fellowship | Ministry of Science and Technology **2025 – 2026**

- Awarded a $500 NST Fellowship under the Physical Science category to support M.Sc. research.

2nd Runner-up, Zelf Hackathon 2.0 **Oct 2024**

- Placed 3rd among 300+ candidates in a QA-focused hackathon, leading to a recruitment interview and subsequent hire.

Merit Scholarship | Jahangirnagar University **2019 – 2025**

- Received $100/year merit scholarships during both B.Sc. and M.Sc. for academic excellence (Top 10%).

Competitive Programming **2019 – 2024**

- Solved 700+ problems on LeetCode, Codeforces, and UVa; participated in national and international programming contests including ICPC 2021, ADA Lovelace NGPC 2021, and NGPC 2021 (ICT Division).

Test Scores & Certifications

IELTS Academic: Overall 7.5 (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0) **Jun 2026**

Fundamental Information Technology Engineer (FE) Examination – Full Passer **Oct 2023**

- Ranked **35th** nationally in the ITEE, a national-level IT engineers' examination governed by IPA, Japan.

Work Experience

Software Quality Assurance Engineer | Zelf AI | Remote Brooklyn, NY, USA | **Dec 2024 – Present**

- Perform root-cause analysis of bugs, audit production NLP features (translation, sentiment, summarization, AI insights) via Google Cloud logs, database queries, and API testing; reported 50+ previously undetected defects.
- Build and maintain automated test suites in Playwright alongside manual functional, regression, and UAT cycles; track issues in Jira, coordinating with developers and stakeholders.

Junior SQA Engineer | Dynamic Solution Innovators Dhaka, Bangladesh | **Apr 2024 – Dec 2024**

Project: KeyPleaz – a US-based app connecting real estate agents instantly.

- Served as the sole QA engineer on the project; authored and managed 500+ test cases in Qase, covering functional, regression, smoke, sanity, and UAT cycles.
- Owned end-to-end testing of high-risk features including in-app chat, calling, and Stripe payment transactions across multiple devices, identifying edge-case failures before release.
- Presented features and demos directly to the US client and coordinated with PM, BA, and developers on user story reviews, ensuring requirements aligned with implementation.

Projects

OBE-Based Result Management System | *Python, Django, Bootstrap, SQLite* [GitHub]

- Built a role-based academic results platform where students view results, teachers analyze courses via auto-generated CO-PO attainment graphs, and administrators monitor performance department-wide.

Konta Niben – an E-commerce System | *MERN, Tailwind CSS, Jest, GitHub Actions* [GitHub]

- Developed a full-stack e-commerce platform following Agile/SDLC practices across 2 sprints, with JSDoc documentation, Jest unit tests, and CI via GitHub Actions.

JU CSE Routinery | *Java, Android SDK, SQLite* [GitHub]

- Built an Android app for organizing CSE department class routines for individual batches.

BoiPoka | *React, Tailwind CSS* [GitHub]

- Developed a front-end bookstore application for browsing and discovering books.

Skills & Technologies

Languages: Python, C++, C, Java, JavaScript, SQL
ML/AI: PyTorch, scikit-learn, OpenCV, NLTK, Hugging Face Transformers, NumPy, Pandas, Matplotlib
Web Development: React, Django, Node.js, REST APIs, HTML5, CSS3, Bootstrap 5, Tailwind CSS
Testing & Automation: Playwright, Cypress, Postman, Qase
Databases: PostgreSQL, MySQL, MongoDB
Tools & Platforms: Git & GitHub, Linux, LaTeX, VS Code, Azure Boards, Jira/Trello, Slack

Extracurricular & Leadership Activities

Founding Member | JU Robotics Club **Feb 2023 – Oct 2024**

- Co-founded the club; built a 6-DoF robotic arm and line follower robots, and competed in inter-university events including Technocrats V.1 (IUBAT).

Publication Secretary | JU Computer Club **Jun 2021 – Apr 2024**

- Promoted from Executive Member to Publication Secretary within a year; managed the club's publication team for nearly 3 years, designing posters and promotional materials for events and recruitment drives.